

Web API : JsonWebToken

1.Create AuthController with JWT Generation

```
public class AuthController : ControllerBase
{
    private string GenerateJSONWebToken(int userId, string userRole)
    {
        var securityKey = new
        SymmetricSecurityKey(Encoding.UTF8.GetBytes("mysuperdupersecretmysuperdupersecret"
        ));

        var credentials = new SigningCredentials(securityKey,
        SecurityAlgorithms.HmacSha256);

        var claims = new List<Claim>
        {
            new Claim(ClaimTypes.Role, userRole),
            new Claim("UserId", userId.ToString())
        };

        var token = new JwtSecurityToken(
            issuer: "mySystem",
            audience: "myUsers",
            claims: claims,
            expires: DateTime.Now.AddMinutes(10),
            signingCredentials: credentials);

        return new JwtSecurityTokenHandler().WriteToken(token);
    }

    // GET: api/auth
    [HttpGet]
```

```

public ActionResult<string> Get()
{
    string token = GenerateJSONWebToken(101, "Admin");
    return Ok(token);
}
}

```

Ouput :

```

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJyYb2xlIjoibGVRtaW4iLCJVc2Vy
SWQqOiOiIjE3MDEiLCJleHAiOiJlE3NTE4NjM2MDAsImZcyI6Im15U3lzdGVtIiw
iYXVkbGoiOiBXLVc2VycyJ9.XXXXXXXXXXXXXXXXXXXXXXXXXX

```

2.Secure EmployeeController with [Authorize]

```

[Route("api/[controller]")]
[ApiController]
[Authorize]
public class EmployeeController : ControllerBase
{
    private readonly List<Employee> _employees;

    public EmployeeController()
    {
        _employees = new List<Employee>
        {
            new Employee
            {
                Id = 1,
                Name = "Alice",
                Salary = 55000,
            }
        }
    }
}

```

```

        Permanent = true,
        Department = Department.IT,
        Skills = new List<Skill> { new Skill { Id = 1, Name = "C#" } },
        DateOfBirth = new DateTime(1990, 5, 12)
    },
    new Employee
    {
        Id = 2,
        Name = "Bob",
        Salary = 48000,
        Permanent = false,
        Department = Department.HR,
        Skills = new List<Skill> { new Skill { Id = 2, Name = "Recruitment" } },
        DateOfBirth = new DateTime(1988, 8, 23)
    }
};
}

```

```

private List<Employee> GetStandardEmployeeList()
{
    return _employees;
}

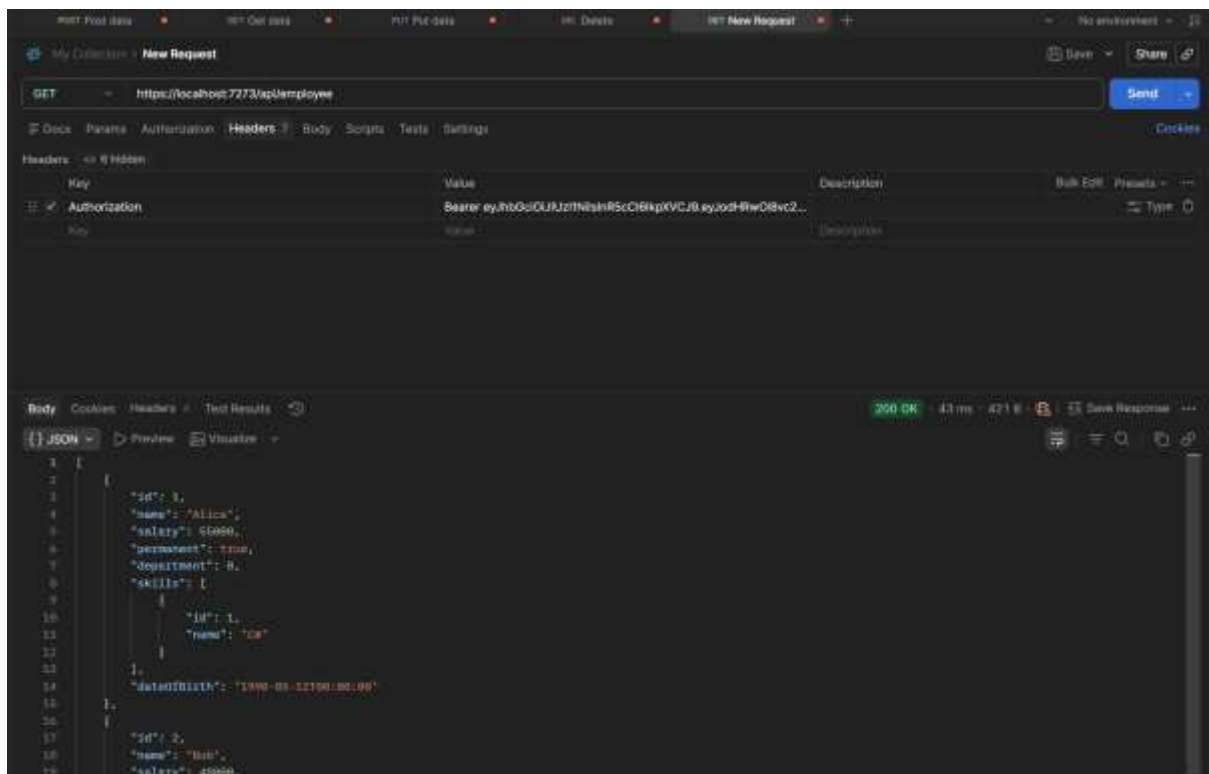
```

```

// GET: api/employee
[HttpGet]
[ProducesResponseType(200)]
[ProducesResponseType(401)]
public ActionResult<List<Employee>> Get()
{
    return Ok(GetStandardEmployeeList()); }
}

```

Output :

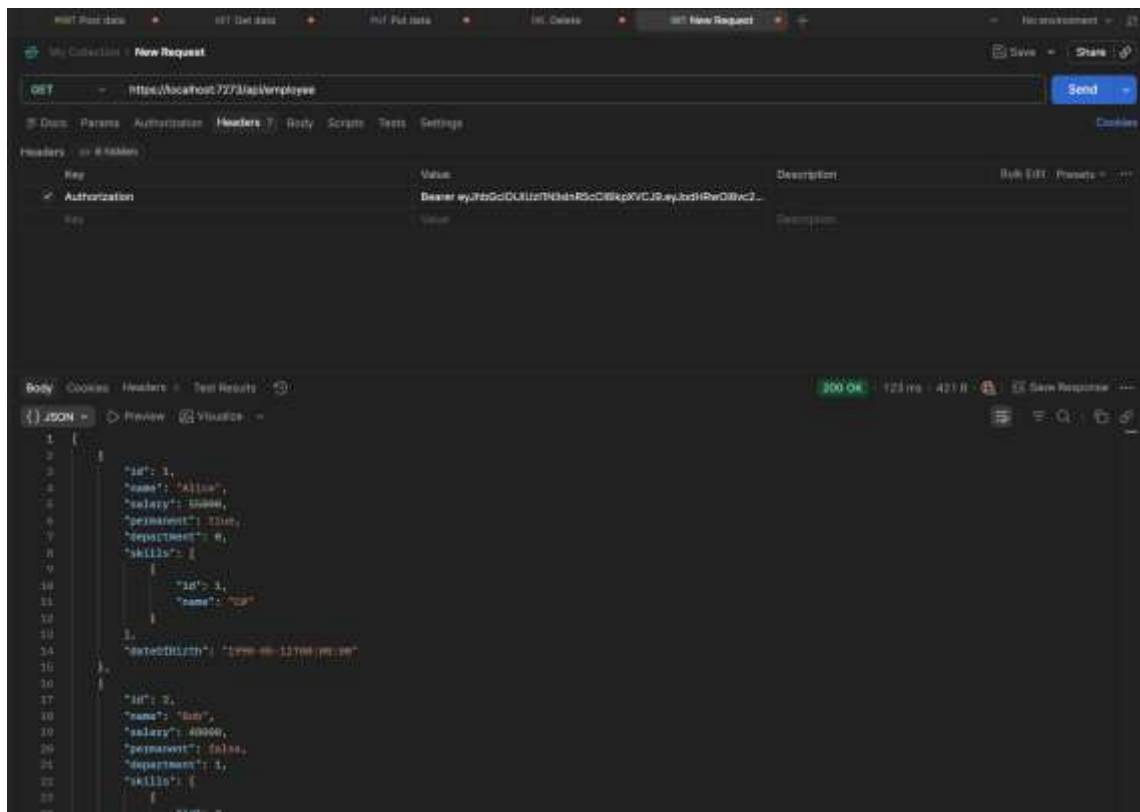


3. Test JWT Expiration :

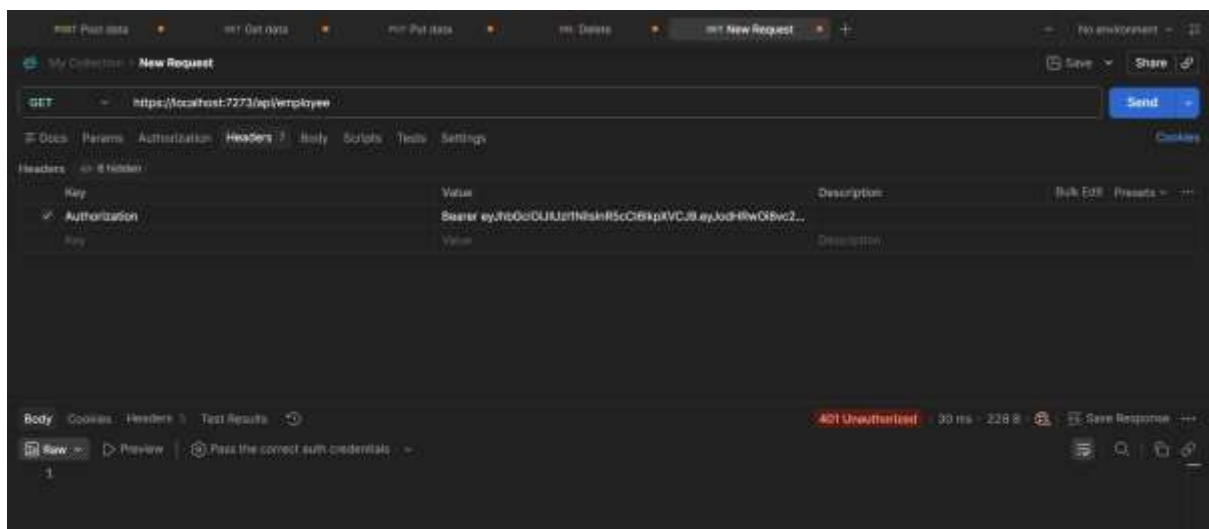
Set –

expires: DateTime.Now.AddMinutes(2)

First using new Token :

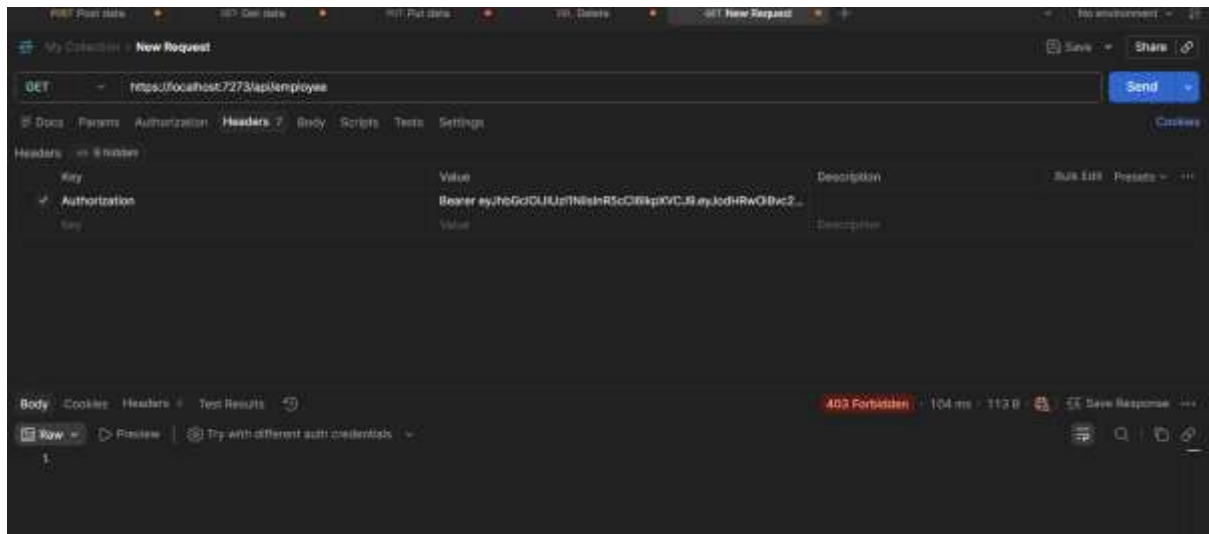


After 2 min , Token Expire :



4.Role-Based Authorization

[Authorize(Roles = "POC")]



After use the , [Authorize(Roles = "POC,Admin")]

